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Macro-Micro Robotic System with a Flexible Continuum Tip for Minimally Invasive Surgery

Interim Report and Project Plan

Abstract

This interim report and project plan is a Thesis A submission that lays the groundwork for improvements to a UR5e based macro–micro robotic system with a flexible continuum tip for minimally invasive surgery (MIS). It consolidates relevant findings from prior work and available literature in relevant fields, and critically analyses how these prior works are related to this thesis. The review explores three technical threads relating to improvements of the UR5e based system. Control improvements for teleoperation, tool exchange mechanisms for modularity, and safety and reliability considerations are reviewed, with a focus on identifying specific gaps that this thesis can address. The report finds clear research opportunities in these areas, specifically around smoothing for teleoperative control, integrated tool exchange validation, and comprehensive safety analysis.

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Nomenclature

A_c	=	tendon/cable cross-sectional area
D	=	backlash deadband
E	=	Young's modulus of tendon/cable material
M_{grasp}	=	estimated distal grasping torque
R_0	=	driving pulley radius
R_5	=	driven pulley radius
R_{eq}	=	equivalent circle radius for tendon/sheath geometry
R_i	=	orientation matrix of segment i
T_i	=	homogeneous transform of segment i
$\Delta\theta_{out}$	=	output side angular backlash/displacement loss
Ω	=	total cumulated curve angle of a tendon/sheath path
θ_{comp}	=	feedforward backlash compensation term
μ	=	friction coefficient (cable/pulley contact)
θ_i	=	local bending angle of segment i
α_i	=	local orientation offset of segment i
$\boldsymbol{p}$	=	rod centerline position in Cosserat formulation
$\boldsymbol{R}$	=	rod orientation in Cosserat formulation
$\boldsymbol{n}$	=	internal force in Cosserat formulation
$\boldsymbol{m}$	=	internal moment in Cosserat formulation
MIS	=	minimally invasive surgery
PCC	=	piecewise constant curvature
RCM	=	remote centre of motion
RMSE	=	root mean square error
UR5e	=	Universal Robots UR5e robotic arm

1 Introduction

1.1 Project Context and Problem Significance

This report aims to extend a previous teleoperated macro–micro surgical robot prototype developed by 2023 UNSW thesis student Lachlan Scott [1]. Scott's project established the viability of such a system and also outlined potential areas of improvement, which this report addresses. Furthermore, new improvement and research possibilities have been identified, such as smoothing the motion of the robotic UR5e arm, the modularity of the system and thus ease of tool exchange, and the safety and reliability of the system for clinical use. Hence, this review of the literature and the thesis question that follows need to recognise this context with regards to Scott's work. This report accordingly defines a focused research question and supporting project plan, and describes work undertaken to ensure feasibility and alignment with the identified knowledge gaps.

2 Literature Review

2.1 Minimally Invasive Surgery (MIS) and Robotic Systems

2.1.1 Historical Progression of Surgery and MIS Robotics

Technology and surgery are inseparable; every surgical advancement or care improvement has been enabled by the parallel development of instruments and tools. From anaesthesia to antiseptics, innovation has allowed surgeons to reduce risk and expand the scope of treatable conditions [2]. In the 19th century, the transition from candlelight to electric lighting allowed for practical internal visualisation, marking the beginning of a major paradigm shift in the surgical field [3]. Electricity also enabled development of various specialised surgical procedures, such as electrocautery for haemostasis and the use of electric drills for orthopaedic surgery [3]. Over time, the focus has shifted towards precision and dexterity, and techniques such as laparoscopy and endoscopy were developed at the beginning of the 20th century, resulting in some of the first minimally invasive surgeries (MIS) [4]. Laparoscopic surgery is a core MIS approach in which instruments are inserted through small abdominal incisions using trocar access ports, rather than a large open incision [5]. Holistically, MIS denotes procedures that use these reduced access pathways to lower tissue trauma and recovery burden relative to open surgery. This trend towards less invasive procedures has continued, and revolutionary robotics like the da Vinci Surgical System became widely adopted in the 1990s and 2000s [6], however there are still challenges to overcome.

2.1.2 Macro–Micro Architecture for Surgical Robotics

The current aspiration of modern surgical robotics is achieving human level dexterity, accuracy, and force control in increasingly small and complex anatomical spaces with limited access and vision. One key technological direction enabling this transition is MIS tooling with a macro–micro architecture. This terminology refers to a robotic system where a larger, more rigid macro manipulator provides positioning and stability, while a smaller, more flexible micro manipulator is attached to the distal end of the macro arm, providing finer control in confined spaces [1].

The macro unit provides larger workspace access and efficiency; however, its larger structure can be prone to deformation and vibration. In contrast, the micro manipulation unit provides the distal dexterity required for manipulation in confined anatomy, where surgical targets often require end effectors in the 3–4 mm class for practical access through narrow channels and natural orifices.

An example of this macro–micro setup is shown in Fig. 1, where a UR5e macro robotic arm is integrated with a tendon-driven continuum micro manipulator [1].

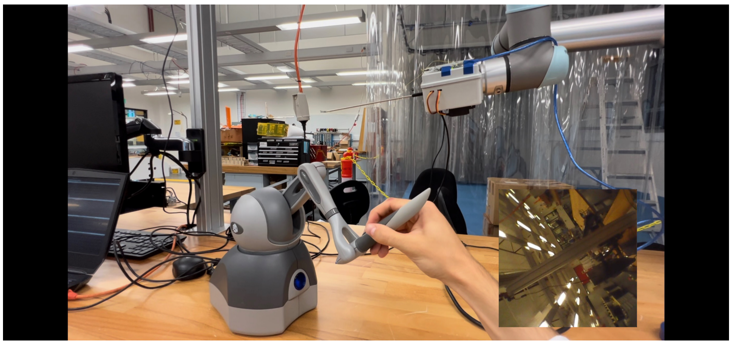


Figure 1: Macro–micro teleoperation system combining a UR5e robotic arm with a continuum tip manipulator [1].

In current snake-like manipulator literature, this micro stage is typically tendon or cable driven to achieve controlled high curvature and orientation control, but the performance relies on the quality of motion transmission, sensing, and control architecture involved [7]. Regarding macro–micro robotics, this local dexterity complements the macro arm's gross positioning capability and could isolate the endpoint from macro stage vibration through coordinated control [1]. A systems level representation of the implementation of this concept is shown in Fig. 2, illuminating the component integration.

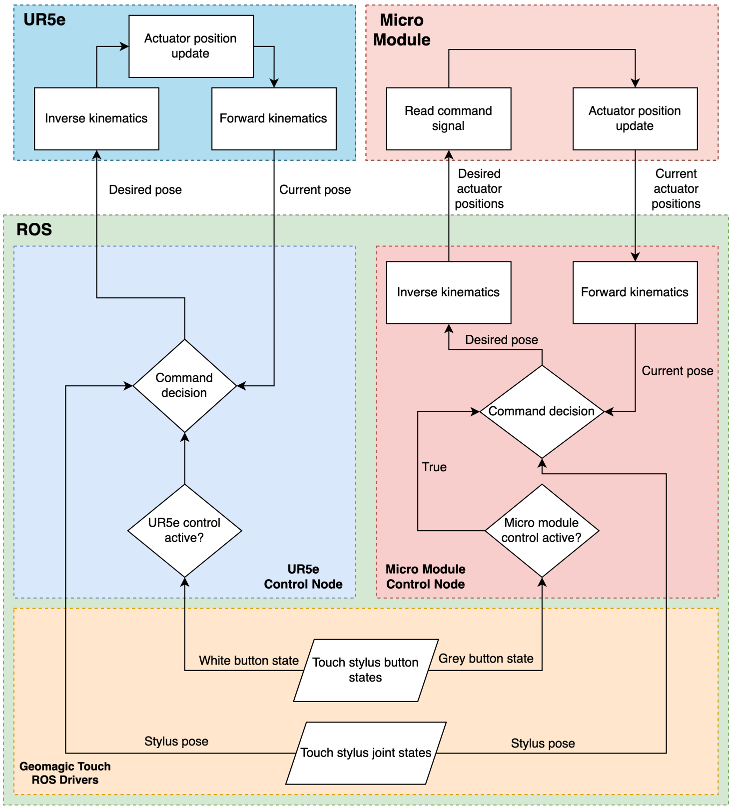


Figure 2: System architecture of a UR5e based macro–micro surgical manipulator [1].

2.1.3 Continuum Manipulators for MIS

Continuum tips are a type of hyper-redundant end effector that can bend continuously along their length. Their high dexterity and slender geometry allow them to navigate tight, curved pathways such as natural orifices and small incisions, where traditional rigid tools may not fit. Tool curvature is generated through differential tension about a central backbone, resulting in multi-DOF distal steering, making these tips suited to macro–micro architectures [7]. For example, the tendon driven continuum tip in Fig. 3 is integrated with a Raven II surgical robot for teleoperation [8].

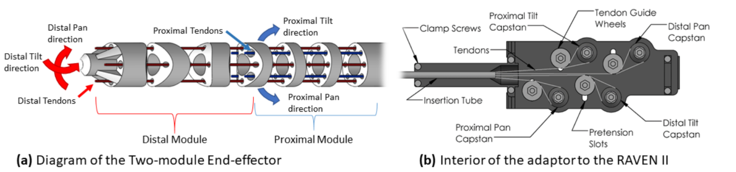


Figure 3: SnakeRaven macro–micro surgical manipulator concept [8].

2.1.4 Cost and Accessibility Considerations

Growing demand and an expanding surgical robotics market has provoked ongoing debate over whether the high cost of integrated platforms such as the da Vinci system limits accessibility. Acquisition costs have been reported up to $4 million AUD, with additional per procedure costs of approximately $1000–$5000 [9]. Despite clear clinical benefits, providers must still evaluate cost effectiveness, especially in low and middle income environments where there are more adoption barriers than the initial cost. Repeated maintenance needs, disposable instrument costs, and ongoing training requirements can prevent usage even after installation [10].

Systemic constraints can also arise from limited infrastructure, workforce training gaps, and the need to prioritise basic surgical capacity over advanced robotic systems [10]. These factors collectively highlight a significant translational gap between technological capability and real world clinical deployment. This literature indicates that scalable, cost effective robotic solutions remain underdeveloped, and thus there is a need for affordable macro–micro systems that combine cheaper hardware with clinically useful teleoperation performance. This aligns with broader advocacy of off-the-shelf hardware and open source software frameworks to reduce cost yet maintain the precision and safety requirements of minimally invasive surgery [1].

2.1.5 UR5e Based Systems and Identified Limitations

The UR5e is a common, economical robotic manipulator that can serve as the macro stage in macro–micro surgical architectures, with reported pricing around $50,000 to $60,000 AUD [11]. Its lower cost and flexibility makes it suitable for integration with custom micro manipulation components. The feasibility of this architecture has been demonstrated in prior thesis work by Scott [1], which implemented a prototype teleoperation pipeline using open source ROS. There, a tendon driven continuum tip was integrated with the macro arm and operated via a haptic interface, following a common snake-like design.

Measurements using electromagnetic tracking demonstrated close pose mirroring under teleoperation, with reported average absolute position errors of 0.0272 mm and 0.0161 mm across trial groups. It is important to note that these values were measured in static pose mirroring tests without external loading, so dynamic tracking and loaded performance remains insufficiently quantified. However, a few practical engineering limitations remain like translational commands inducing vibration in the integrated system, which negatively affects motion stability and repeatability. Furthermore, the current modular workflow requires manual exchange of tendon-driven end effectors, resulting in increased setup time and reduced operational efficiency.

These limitations reveal a key gap in existing UR5e based macro–micro systems: while feasibility and precision have been demonstrated, improvements in vibration suppression, system robustness, and workflow integration are required to be clinically viable regarding safety and usability [1], [7], [8].

2.2 Teleoperation and Control in Surgical Robotics

2.2.1 Modelling Frameworks for Continuum Surgical Robotics

To explore control and modularity claims on a geometric basis, kinematic, piecewise constant curvature (PCC), and mechanical modelling is employed. At the kinematic level, each segment transform can be expressed as

$$T_i = \begin{bmatrix} R_i & p_i \\ \mathbf{0}^\top & 1 \end{bmatrix}, \tag{1}$$

which maps segment states to distal pose and forms a computationally efficient baseline for real time teleoperation [12], [13]. However, it does not capture load dependent deformation.

For tendon-driven systems, PCC models provide a relevant mapping:

$$T_i^{i+1} = \mathrm{rot}(Z_i, \alpha_i)\,\mathrm{trans}(Z_i, H_i)\,\mathrm{trans}(X_i, W_i)\,\mathrm{rot}(Y_i, \theta_i)\,\mathrm{rot}(Z_i, -\alpha_i), \tag{2}$$

and are widely used due to their balance between efficiency and geometric fidelity [13], [14].

When compliance, external loading, and routing effects dominate, mechanical models such as Cosserat rods better encapsulate continuum dynamics, incorporating distributed forces, moments, and deformation along the manipulator.

No single modelling layer is sufficient for surgical workflows: kinematic models are efficient but approximate, while mechanics based models are more expressive but computationally demanding. This tradeoff encourages mixed modelling and compensation strategies in teleoperated systems. Within this context, the control literature can be interpreted across linked layers, from elementary transmission effects to high level shared control, as summarised in Fig. 4.

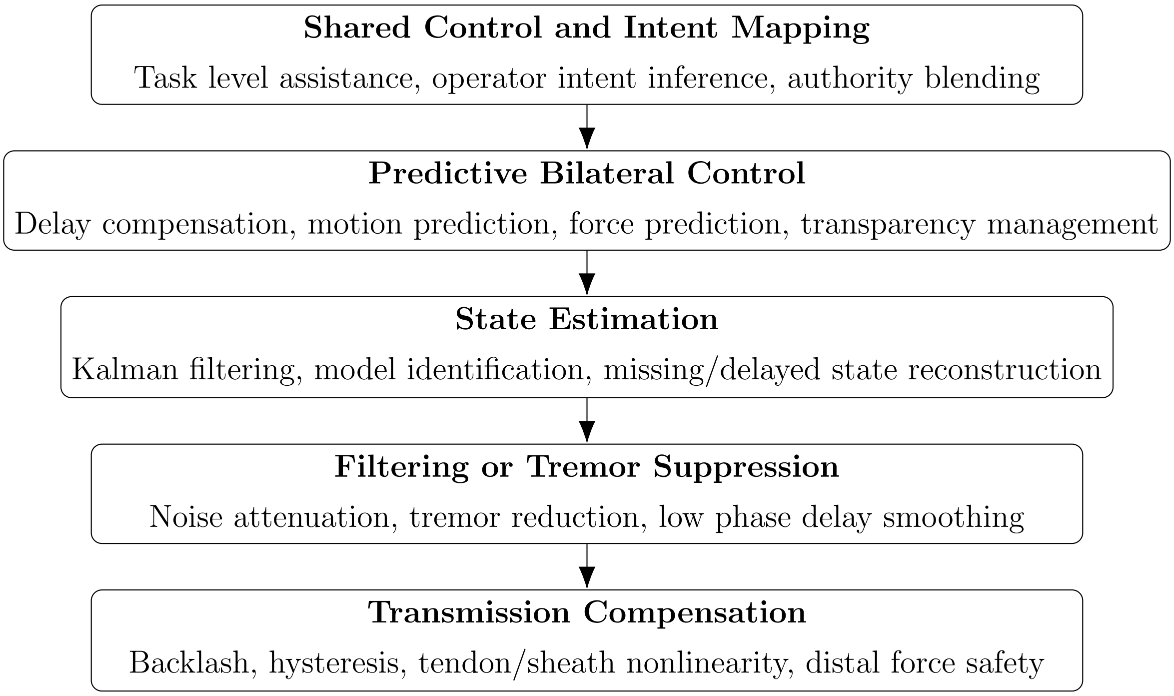


Figure 4: Linked control layers relevant to teleoperation in macro–micro surgical robotics

2.2.2 Tremor Suppression and Delay Robust Interaction

Human physiological tremor is a significant control issue in robotic MIS because involuntary operator motion is captured in the teleoperation loop and may appear as vibration at the surgical instrument tip. Sang, Yang, Liu, *et al.* [15] suggest the need to balance competing goals: strong attenuation of involuntary motion and minimal phase delay to avoid latency. They combine a zero phase prefilter with adaptive Kalman estimation to improve tremor estimation and compensation in real time, and successfully lowered error and reduced tip vibration. This is directly relevant to the present report as it frames tremor control not as an isolated signal processing problem, but as a coupled teleoperation and tracking problem where compensation quality, delay, and physical dynamics are all relevant [15].

When haptic feedback is included, system latency can bottleneck teleoperation. Batty, Ehrampoosh, Shirinzadeh, *et al.* [16] show a fundamental transparency versus stability tradeoff, where increasing force fidelity can reduce stability under delay, while aggressive stabilisation degrades operator feedback. Consequently, they estimate the environment online, and master feedback is generated through this prediction rather than delayed direct force reflection. Their human in the loop palpation experiments demonstrate that this predictive approach substantially reduces force prediction error, and improves teleoperation transparency under delay. This indicates that latency management is mainly a control architecture problem, requiring integration of prediction, filtering, and system modelling to maintain stability and responsiveness [16].

2.2.3 State Estimation and Predictive Bilateral Architectures

Filtering is not optional in surgical teleoperation, rather a requirement to maintain stable and trustworthy state estimates despite noisy measurements and imperfect network transport. Lashari, Batayneh, and Khokhar [17] show that a Kalman filter paired with state-space models can remain computationally efficient and still estimate manipulator position with high tolerance for delay, jitter, and packet loss. By developing models directly from surgical data, this approach prevents reliance on idealised system assumptions while still smoothing disturbances and preserving continuity. Unfortunately, validation is primarily based on simulation and dataset driven conditions, and generalising to real surgical environments remains difficult [17].

Predictive bilateral control is also emerging as a practical strategy for continuum and snake-like teleoperation under delay. Ebrahimian, Pourmokhtari, Ghiyasi, *et al.* [18] propose an online architecture that combines master sideforce prediction with operator motion prediction on the slave side, which aims to reduce communication lag and preserve position synchronisation and haptics. This addresses the same core bottleneck as latency and filtering studies, but at the controller level, meaning delay compensation is handled through coordinated forward and backward prediction, rather than being one sided. Their simulation results show strong delay bypass for known constant delays in snake-like robot teleoperation, however performance is demonstrated mainly in simulation with structured assumptions that may not fully capture higher complexity interactions [18]. Combined, it is apparent that robust teleoperation under delay requires both reliable state estimation and predictive control, reinforcing the need for integrated architectures rather than isolated solutions.

2.2.4 Shared Control, Immersion, and Haptic Force Awareness

Shared control combines user input with autonomous assistance, and thus introduces the challenge of truly mapping teleoperated motion to user intent. Bowman, Zhang, and Zhang [19] argue that just direct motion mapping is insufficient for robust telemanipulation when the human and robot hands are physically dissimilar, as geometric tracking does not always correlate with overall task success. Their intent based shared control framework instead blends operator following behaviour with autonomous task assistance using inferred intent probabilities and adaptive arbitration weights. Critically, this contribution formalises how authority should shift based on hand structure discrepancy and task constraints, which is directly relevant to macro–micro MIS. This furthers the view that task completion and perceived control are the success criteria, rather than tracking endpoint error alone [19]. Unfortunately, validation is limited to controlled tasks, and transferability to complex surgical scenarios remains uncertain.

Recently, immersive visualisation (VR) and haptic feedback are becoming more relevant in surgical teleoperation [20]. Improved viewpoint control and force feedback enhance situational awareness and reduce applied force during manipulation, with reported reductions in tissue interaction forces in emerging surgical systems. This suggests that perception and force regulation are coupled design variables, rather than independent interface features.

A key limitation of this evidence is heterogeneity across platforms, systems and evaluation methods, limiting the usefulness of direct comparison. Therefore, this should only be interpreted as trend evidence for evaluation criteria such as situational awareness, workload, and force moderation, rather than definitive performance benchmarks.

2.2.5 Transmission Nonlinearity and Distal Force Safety

Transmission backlash in cable and pulley drives is directly relevant to continuum tips because direction reversals create delay and transition phases in which distal motion and torque do not track commands directly. Zou, Li, and Huang [21] model this explicitly feeding forward output backlash, $\theta_{comp} = \Delta\theta_{out} R_5/R_0$, together with grasping torque estimation and saturation limiting, $M_{grasp} = M_{out} - M_{fric}$. This aims to reduce tracking error and limit unsafe force, and their results show meaningful improvements in position and force control of a grasper. Position error is reduced to approximately 1.03° mean (3.49° max), and grasping torque tracking error is reported below 0.025 N·m. Nonetheless, residual error around direction reversal remains, proving accuracy is extremely sensitive to parameter identification and true system friction [21].

Tendon/sheath studies show similar hysteresis effects under variable shape conditions. Hong, Hong, Kim, *et al.* [22] model and compensate route dependent deadband using feedforward calibration, achieving substantial tracking improvement (RMSE reduced from 1.31 mm to 0.19 mm). However, performance remains dependent on reliable calibration and stable geometric conditions [22].

Together, it seems transmission nonlinearity remains a critical limitation in tendon driven surgical manipulators, particularly under variable routing and loading conditions. While feedforward compensation is a common approach to significantly reduce tracking and force errors, performance is still dependent on accurate parameter identification and stable system geometry. These findings establish clear performance requirements for low cost macro–micro platforms and provide a direct relation to modular continuum hardware discussions. The relationship between these control problems within the overall teleoperation architecture is visualised in Fig. 5.

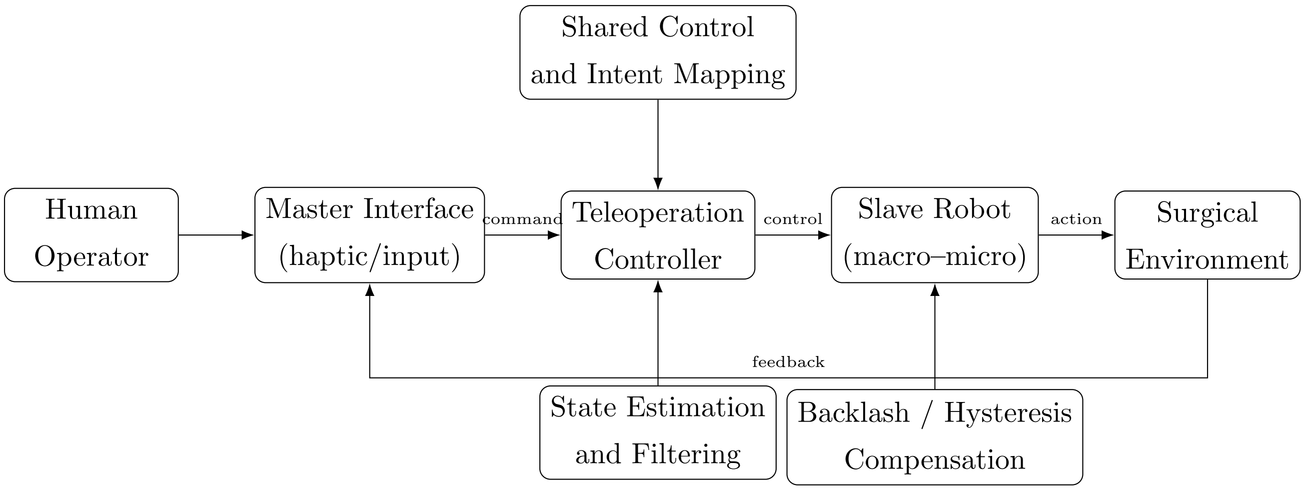


Figure 5: Simplified teleoperation architecture for macro–micro surgical robotics. The reviewed literature addresses different parts of this loop, including state estimation, shared control, predictive control, and transmission compensation.

2.3 Modularity and Tendon-Driven Continuum Manipulators

2.3.1 Monolithic vs Modular Distal Architectures

A monolithic distal architecture is characterised by a single integrated tool assembly where changing function require a slow teardown or disassembly, rather than a rapid tool exchange. The practical consequences of this architecture distinction regarding workflow include longer swap times [23], re-tensioning and recalibration after replacement, interruption of the operative sequence, and reduced intraoperative adaptability when task requirements change during the procedure. These constraints align with broader integration issues seen in modular end effector software and hardware pipelines [24], as many modularity solutions introduce non-trivial changeover overhead.

Internal platform observations in prior local UR5e continuum tip implementations also indicate that tool replacement is slow and introduces practical workflow friction in achieving repeatable setup after each swap [1]. Thus, evidence indicates that architectural modularity alone is insufficient unless exchange time and post swap recalibration costs are explicitly addressed.

2.3.2 Tool Exchange Mechanisms

Tool exchange is necessary because laparoscopic workflows require multiple functional instruments during a single procedure, yet only a limited subset can be mounted simultaneously on the robotic arm [5]. The resulting recurring intraoperative transition can interrupt surgical flow and, depending on the swap method, introduce further coordination variability.

2.3.3 Rail Guided and Cartridge Based Exchange

The most common swap mechanisms are rail guided, cartridge based exchange approaches. Ko, Kim, Lim, *et al.* [5] analyse current sliding exchange practices and report an automated replacement architecture that preserves trocar and remote centre of motion (RCM) constraints via linear motion. Their prototype completed a full exchange in 15 s for a 3.36 kg device, down from an initial 27 s cycle, indicating feasible automation under laparoscopic geometric constraints.

2.3.4 Reconfigurable and Multifunctional Instruments

Alternative strategies aim to reduce exchange frequency rather than improve exchange interfaces directly. Mocellin, Pagliarani, Gamberini, *et al.* [25] present OriGrasp as a reconfigurable multifunctional end effector, that has a 4 N pinch force and bowel pulling or lifting forces that fall in consistent atraumatic ranges. Thus, multifunctionality can be a useful partial mitigation of overhead for selected abdominal tasks, but current evidence still remains narrow and does not demonstrate that a single reconfigurable device can replace the broader set of instruments required across other procedures.

2.3.5 Limitations of Alternative Exchange Mechanisms

Mechanism diversity beyond rail guided exchange remains under validated in the reviewed evidence base. Direct surgical evidence is limited for bayonet twist-lock, cam/latch, and cartridge head variants beyond linear sliding, as well as kinematic coupling assisted docking with active preload. Reported gaps are not purely mechanical as tolerance sensitivity, sterilisation robustness, and post reconnection calibration all remain insufficiently quantified across repeated swap cycles.

2.3.6 Post Swap Control Recovery and Tension Management

Control evidence indicates that docking success alone is insufficient for viable exchange. Cable/pulley and tendon/sheath studies report backlash, hysteresis, and force estimation sensitivity to pretension, friction, and routing geometry [21], [22]. In particular, Hong,

Hong, Kim, *et al.* [22] show geometry dependent deadband behaviour and report RMSE reduction from 1.31 mm to 0.19 mm with feedforward compensation after calibration, implying that post swap recovery may require both re-tensioning and state updates when routing conditions change.

For post swap tension management, Singh and Khatait [26] demonstrate an RCM tension adjustment mechanism that enables independent cable pretension tuning with invariant routed tendon length. However, this work does not address quick change end effector interfaces or repeated docking performance in surgical workflows. A comparative summary of tool exchange strategies is provided in Table 1.

Criterion	**Rail guided sliding**	**Quick-lock / bayonet**	**Reconfigurable tool**
Mechanism concept	Linear cartridge style or rail guided insertion and removal	Rotational locking, latch based docking, or compact quick coupling interface	Single distal tool changes functional mode instead of being physically exchanged
Workflow benefit	Structured and repeatable alignment during exchange; compatible with trocar / RCM constraints	Potentially compact and rapid attachment with fewer gross insertion steps	Reduces exchange frequency by combining multiple functions into one tool
Main limitation	Still requires reliable docking, reconnection, and post swap recovery	Direct evidence for repeatability, sterilisation robustness, and recalibration remain limited	Typically validated only for narrow task classes; unlikely to replace the full instrument set
Evidence base	Strongest direct exchange evidence in the reviewed literature	Sparse direct surgical validation and weak comparative evidence	Promising prototype/task specific evidence, but limited generality across procedures
Validation maturity	High	Low	Moderate

Table 1: Comparison of tool exchange strategy families relevant to modular tendon-driven surgical manipulators

2.3.7 Critical Assessment and Identified Knowledge Gap

The literature indicates a clear hierarchy of maturity: (i) well established rail guided exchange, (ii) promising but task constrained multifunctional tools, and (iii) limited evidence for alternative docking and locking mechanisms. The key gap is the lack of integrated validation of mechanism and control recovery following series of tool exchanges. In particular, exchange time, docking repeatability, post swap backlash and deadband, retensioning effort, and distal force safety must be evaluated jointly under realistic routing and sterilisation conditions.

2.4 System Safety and Reliability Considerations

2.4.1 Safety Constraint Model and Hazard Taxonomy

Safety in macro–micro MIS involves constrained execution within mechanical, control, and workflow limits. In minimally invasive surgery, tools are introduced through a *trocar*, a fixed entry port that enforces a pivoting constraint on the instrument shaft. This is enforced as an RCM constraint, requiring zero velocity at the entry point [12], [13].

Kinematically, trocar RCM constraint enforcement can be represented by

$$\mathbf{v}_{RCM} = J_{RCM}\dot{\mathbf{q}} = \mathbf{0} \tag{3}$$

and in constrained form

$$\dot{\mathbf{x}}_e = J_c\dot{\mathbf{q}}_I, \quad J_c = J_e \begin{bmatrix} I_n \\ -J_{II}^{-1}J_I \end{bmatrix} \tag{4}$$

which enforces motion consistency under kinematic constraints [13].

For moving trocar conditions, such as soft tissue interaction or external disturbance, the constraint can be extended to

$$\dot{\mathbf{q}}_{II} = J_{II}^{-1}\left(\mathbf{v}_{trc} - J_I\dot{\mathbf{q}}_I\right) \tag{5}$$

preserving explicit constraint compensation under disturbance.

These constraints define one safety layer (constraint violation). Additional risks include distal force overshoot, device malfunction, post swap state mismatch, delayed recovery, and degradation from repeated reprocessing.

2.4.2 Error Detection, Frequency, and Clinical Consequence

Rajih, Tholomier, Cormier, *et al.* [27] provide useful evidence on practical malfunction detection, as errors were captured through da Vinci Si event logs, time stamped, and classified as recoverable versus unrecoverable across 1228 procedures. Note that safety analysis depends on what is actually detected and recorded, not just what fails physically. Their workflow separates events by timing and subsystem category, creating a useful template for future macro–micro logging: event source, recoverability, corrective action, and impact on operative flow [27]. They report an overall malfunction rate of 4.97% (61/1228), with most errors recoverable (46/61). Common faults include arm output limits and communication errors, while serious outcomes were rare, with no case abortion, conversion, injury, or death reported. This suggests that malfunctions are not uncommon but are typically low consequence when recovery mechanisms are effective [27].

However, residual risk remains dependent on detection latency, context, and available intervention. To maintain explicit risk representation, this review considers occurrence probability and risk priority metrics:

$$\hat{p}_i = \frac{N_i}{N_{tot}}, \qquad RPN_i = S_i O_i D_i, \tag{6}$$

where $\hat{p}_i$ denotes event frequency and RPN_i combines severity, occurrence, and detectability.

To make this framework operational, Table 2 applies the S, O, and D ratings to representative hazards identified in the reviewed literature.

Hazard event	S_i	O_i	D_i	RPN_i
Distal force overshoot during teleoperation	4	2	3	24
RCM constraint violation near trocar interface	5	2	3	30
Post swap miscalibration causing tip offset	3	3	3	27
Communication dropout during critical manoeuvre	4	2	2	16
Delayed fault detection and alarm response	4	3	4	48

Table 2: Illustrative hazard prioritisation using $RPN_i = S_i O_i D_i$ for representative macro–micro surgical workflow events.

2.4.3 Preventable Versus Non-Preventable Faults

Rajih, Tholomier, Cormier, *et al.* [27] also distinguish between preventable execution faults and less predictable electronic anomalies. Collisions and jerky motions are identified as a major recoverable class linked to operator technique, whereas communication board faults typically require manufacturer intervention. This distinction implies that mitigation strategies must differ: port and arm planning, as well as operator training, can reduce preventable faults, whereas non-preventable faults require rapid diagnosis, robust recovery, and fallback workflows [27].

2.4.4 Lifecycle Reliability: Reconfiguration, Wear, and Reprocessing

Short term controller performance is not sufficient to ensure safety. Sadeghian, Zokaei, and Hadian Jazi [28] demonstrate feasible constrained tracking ($\pm 0.4\,\mathrm{mm}$), but this does not address prolonged workflow stress or repeated reconfiguration effects.

Saito, Yasuhara, Murakoshi, *et al.* [29] show that residual protein levels remain substantially higher in robotic instruments than in conventional instruments after cleaning (robotic: 650, 550, 530 μg/instrument vs. ordinary: 16, 17, 17 μg/instrument; $P < 0.0001$). The practical implication is that sterilisation and reprocessing must be treated as tracked reliability variables when repeated tool exchange and reuse are core to platform operation.

2.4.5 Integrated Safety Gap and Evaluation Endpoints

The supported safety gap is the lack of integrated validation across *constraint control, malfunction detection and recovery, and lifecycle reprocessing reliability* within a single workflow. Accordingly, evaluation should use unified metrics, including RCM error bounds, recovery time, post swap repeatability, residual backlash, distal force limits, and contamination levels across repeated cycles. These metrics are conceptually aligned with formal medical device frameworks, including ISO 14971 risk management, IEC 60601 electrical safety and essential performance requirements, and IEC 62304 software lifecycle expectations [30]–[32]. This review does not have full text access to these standards and they are referenced only to structure measurable and auditable safety endpoint goals.

2.5 Synthesis of Literature and Knowledge Gaps

Continuum tip hardware studies support micro dexterity in constrained anatomy, while teleoperation and control studies show that tremor, delay, estimation uncertainty, and intent mapping are all critical to performance. Malfunction events are shown to be significant but generally manageable, and accessibility studies introduce a practical constraint on performance under realistic cost, maintenance, and workforce limits, not only in high resource laboratory settings.

However, these areas have been validated in isolation. Kinematic and compensation studies assume controlled routing and loading; teleoperation studies assume simplified communication conditions; malfunction studies focus on event reporting rather than causal mechanisms; and lifecycle studies quantify reprocessing effects without linking them to control state recovery. This fragmentation limits understanding of end-to-end workflow reliability.

The relationship between subsystems is direct as UR5e macro stage provides accessible gross positioning, while the continuum distal stage provides local dexterity. The interface between them can introduce coupled failure modes that include vibration transmission from the macro arm, backlash and hysteresis in tendon-driven actuation, post swap recalibration burden following tool exchange, and safety assurance under disturbance. The research problem is therefore not limited to improving a single controller or mechanism, but to determining whether a low cost macro–micro architecture can maintain stable, smooth, and safe operation when control, mechanics, and reconfiguration are exercised together.

Based on this synthesis, four principal gaps emerge. First, there is limited evidence on how macro stage dynamics, such as UR5e vibration and compliance, propagate to distal motion, and how filtering and control strategies can achieve smooth, delay robust teleoperation without degrading responsiveness. Second, rapid and repeatable tool exchange mechanisms remain under validated, particularly with respect to maintaining consistent calibration, tensioning, and transmission behaviour across repeated swaps. Third, safety in operation is insufficiently integrated, with limited evidence linking constraint enforcement, fault detection and recovery, and force regulation to task level outcomes under realistic conditions. Finally, there is a lack of unified experimental frameworks that evaluate control performance, mechanical reliability, and reconfiguration effects together using clinically meaningful and repeatable metrics.

2.5.1 Remaining Evidence Gaps and Next Directions

Beyond this report, several research directions would strengthen translation from laboratory validation to clinically useful systems. While shape sensing has been demonstrated in continuum robots, integration with low cost macro–micro platforms remains sparse; distal force instrumentation could likewise tighten observability where packaging and cost permit. Second, learning assisted controllers could be investigated to adapt filter and control parameters online across different tools and routing geometries while preserving safety bounds.

Another priority is validation across lifecycle and governance constraints. Multi session studies should quantify drift, recalibration burden, and reprocessing effects over repeated exchanges, while future work should map laboratory evidence into pre clinical verification plans that are explicitly linked to risk management and software assurance frameworks.

3 Research Question and Project Plan

3.1 Research Question

Based on the reviewed literature, this thesis addresses the following research question:

> *Can a UR5e based macro–micro robotic system satisfy predefined performance thresholds under controlled laboratory conditions by simultaneously achieving: (i) at least 30% reduction in distal tip vibration amplitude in the peak frequency band relative to an unfiltered baseline, (ii) median end effector exchange cycle time of* ≤ 20 *s with post swap positional repeatability error of* ≤ 1.0 *mm, and (iii) RCM tracking error of* ≤ 1.0 *mm without increasing command to motion latency beyond 50 ms?*

It is hypothesised that integrating zero-phase smoothing control, repeatable end effector exchange, and constraint aware monitoring will satisfy all three thresholds above and failure to meet any one threshold constitutes rejection of the hypothesis. While macro–micro architectures have demonstrated high positional accuracy and feasibility [1], they remain affected by macro stage vibration, manual and time consuming end effector exchange, and fragmented approaches to safety evaluation. The literature further shows that teleoperation performance, modularity, and safety are typically studied in isolation, limiting the ability to assess overall workflow reliability. Accordingly, this thesis focuses on improving and testing the system at a systems level, rather than proposing isolated algorithmic or mechanical improvements.

The 30% vibration target is intended as a meaningful but achievable improvement threshold for a low cost platform rather than an optimum. The 20 s exchange target is deliberately more lenient than the 15 s automated exchange reported by Ko et al. [5], because this project prioritises a repeatable workflow over full automation. The 1.0 mm RCM target keeps the constraint metric in the sub millimetre range while remaining less stringent than the constrained tracking performance reported in prior work [28].

3.2 Project Plan

The project is organised as a progressive experimental workflow, in which teleoperation control, modular exchange, and safety are evaluated sequentially and then together. These are not independent tasks, rather they comprise a single experiment. Performance will be evaluated using quantitative metrics including motion smoothness, effector swap time and accuracy, and adherence to constraint and safety bounds.

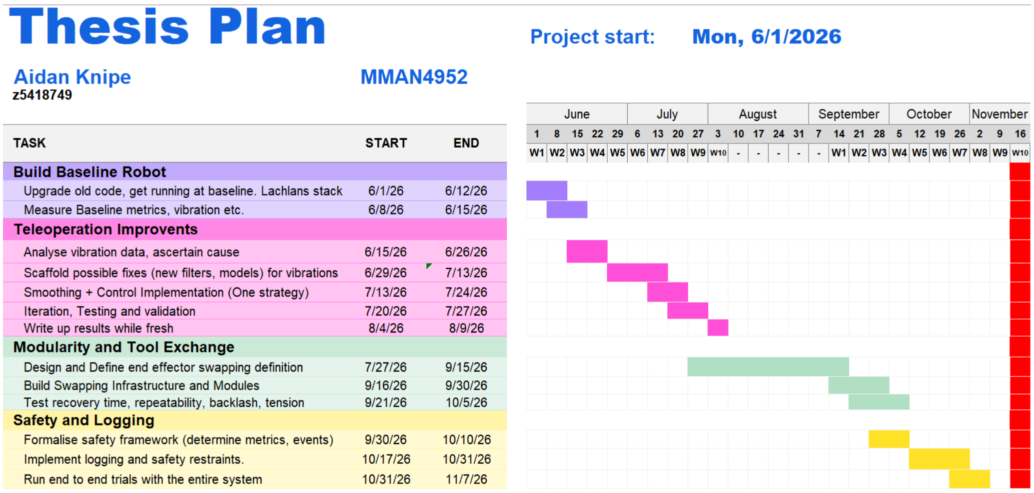
Regarding teleoperative control, prior studies show that tremor, delay, and vibration transmission significantly affect precision and usability [15], [16]. In this project, effort is therefore directed towards establishing a stable baseline system and implementing a practical smoothing strategy to improve motion quality without introducing excessive latency. This work also serves as the foundation for all subsequent evaluation, as reliable teleoperation is required before modularity and safety can be meaningfully assessed.

The second component focuses on modular end effector exchange. While tool exchange is necessary in laparoscopic workflows, the literature highlights a lack of validation of post swap behaviour, particularly with respect to calibration, tensioning, and transmission effects [5], [22]. Rather than developing a complex new mechanism, this project prioritises a repeatable exchange and recovery workflow, with the goal of measuring the impact of swapping on system performance. This aligns with the identified gap that mechanism design alone is insufficient without evaluating post swap recovery.

The third component addresses safety in an operational context. Existing studies demonstrate that malfunctions are not uncommon but are typically recoverable when appropriate detection and response mechanisms are present [27]. However, these are rarely linked to a macro–micro system, so this project therefore evaluates safety using integrated metrics, including constraint adherence, recovery time, and repeatability across operation cycles, rather than relying on isolated fault statistics.

The scope of the project is constrained by the available time (approximately 200 hours across Terms 2 and 3) and the use of an existing UR5e based platform. This means a focus on implementing and validating representative solutions has been adopted, rather than developing completely new hardware or control frameworks. This ensures that the project remains feasible while still addressing the key gaps identified in the literature. The project relies on access to the UR5e platform, ROS2 based control infrastructure, and completion of required laboratory training and coursework.

A detailed timeline is provided in Fig. 6. The plan is organised across two terms, with Term 2 focused on system setup and teleoperation control, and Term 3 focused on modular exchange evaluation, safety testing, and integrated experiments. The Gantt chart captures the specific task breakdown, dependencies, and time allocation.



Thesis Plan

Project start: Mon, 6/1/2026

Aidan Knipe MMAN4952
z5418749

TASK	START	END
Build Baseline Robot		
Upgrade old code, get running at baseline. Lachlans stack	6/1/26	6/12/26
Measure Baseline metrics, vibration etc.	6/8/26	6/15/26
Teleoperation Improvents		
Analyse vibration data, ascertain cause	6/15/26	6/26/26
Scaffold possible fixes (new filters, models) for vibrations	6/29/26	7/13/26
Smoothing + Control Implementation (One strategy)	7/13/26	7/24/26
Iteration, Testing and validation	7/20/26	7/27/26
Write up results while fresh	8/4/26	8/9/26
Modularity and Tool Exchange		
Design and Define end effector swapping definition	7/27/26	9/15/26
Build Swapping Infrastructure and Modules	9/16/26	9/30/26
Test recovery time, repeatability, backlash, tension	9/21/26	10/5/26
Safety and Logging		
Formalise safety framework (determine metrics, events)	9/30/26	10/10/26
Implement logging and safety restraints.	10/17/26	10/31/26
Run end to end trials with the entire system	10/31/26	11/7/26

Figure 6: Project timeline and task allocation across Terms 2 and 3.

To support planning and identify potential risks, a mind map has been developed that includes task dependencies, alternative approaches, and anticipated failure points (Fig. 7). In particular, it identifies areas where scope may need to be reduced, such as simplifying the exchange mechanism or limiting the complexity of control strategies, ensuring that meaningful results can still be obtained within the available timeframe. Key contingencies include simplifying the control strategy if integration proves difficult, and validating a manual but repeatable exchange workflow if hardware modifications are not feasible. These ensure that core evaluation metrics can be obtained within the project timeframe.

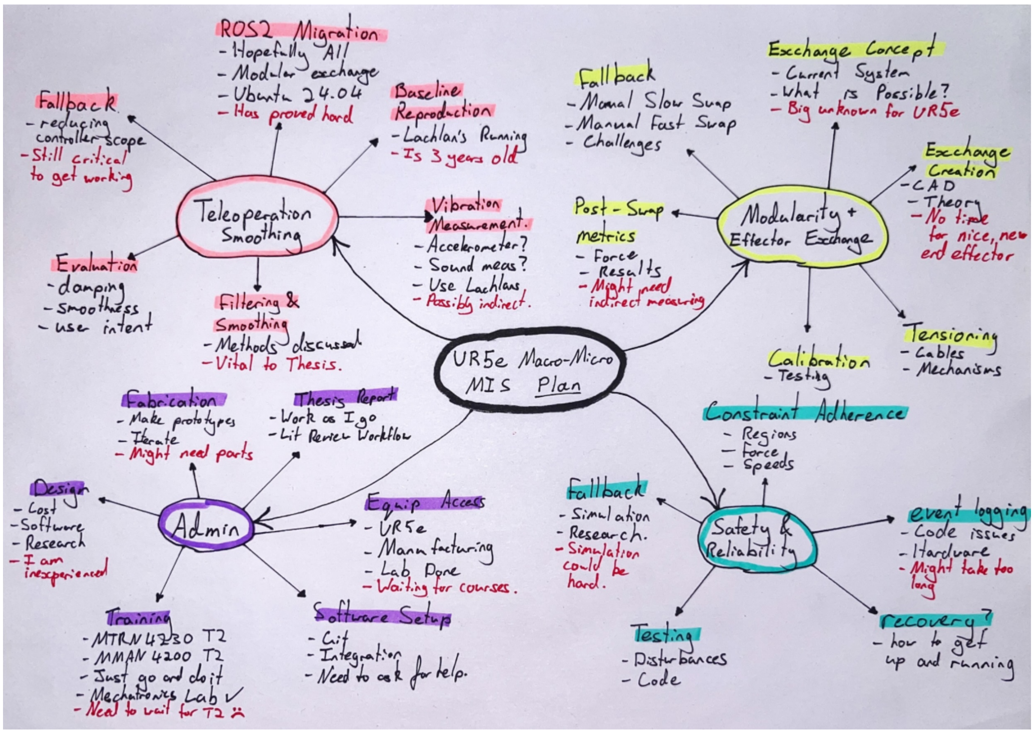


Figure 7: Mind map used fo project planning, including risks and alternative approaches.

Overall, the project plan is intentionally structured to reflect the narrative developed in the literature review. By focusing on teleoperation smoothing, modular exchange recovery, and safety evaluation within a single system, the work aims to provide a coherent assessment of macro–micro robotic performance under realistic conditions, rather than treating each component in isolation.

4 Project Dependent Preparations

Preparatory work has been undertaken to ensure that the project can proceed without major revision, focusing on software upskilling, system access, and early risk identification. These preparations align with the key weaknesses identified in the literature, particularly the reliance on robust software integration, reliable teleoperation infrastructure, and repeatable experimental workflows.

4.1 Technical Preparation and Upskilling

A working Ubuntu 24.04 development environment has been established and ROS2 has been successfully installed and tested. Basic publisher–subscriber communication has been verified using demonstration nodes, confirming correct system setup and message passing within the ROS2 framework.

This preparation directly addresses a key project dependency, as prior work was implemented in ROS and will require adaptation to ROS2. Early inspection of the codebase and system architecture from Scott [1] has provided familiarity with the macro–micro teleoperation pipeline, reducing integration uncertainty and establishing a baseline understanding of system structure.

Formal upskilling is also in progress through enrolment in MTRN4230, which provides training in robot control and ROS based systems and is required for UR5e access. In addition, taking MMAN4200 (Additive Manufacturing) supports skills for rapid iteration of modular end effector components.

4.2 System and Resource Readiness

Mechatronics laboratory induction has been completed, and the UR5e system and existing continuum end effector have been physically inspected. This provides practical understanding of the platform and confirms system availability.

Access procedures for the robot have been established, with experimental work to commence following completion of required training. The project also benefits from inheriting an existing macro–micro system, reducing development risk and allowing focus on performance improvement over construction.

4.3 Preliminary Observations and Risk Mitigation

Early preparation has identified several key risks consistent with the literature. Software integration, particularly migration from ROS to ROS2, remains non-trivial but has been partially mitigated through early environment setup and code familiarisation. Hardware access is dependent on training and lab availability, which is being addressed through course enrolment and established booking procedures. System complexity in macro–micro integration is mitigated by leveraging an existing validated platform rather than developing new hardware.

These preparations indicate that the project is ready to proceed, with key technical dependencies established and major risks identified at an early stage.

5 Conclusion

The reviewed literature revealed three key development areas. Macro–micro and tendon-driven continuum systems are well suited to MIS dexterity, but remain sensitive to vibration, modelling error, and teleoperation instability. Control strategies improve performance, yet are typically validated under constrained conditions. Accessibility is further limited by the full system burden rather than capital cost alone.

The primary gap is therefore not isolated improvements, but integrated validation of system level performance. In particular, there is limited evidence linking teleoperation smoothness, modular end effector exchange, and safety critical reliability within a single, repeatable workflow.

This motivates a systems level investigation of a UR5e based macro–micro platform, where control, modularity, and safety are evaluated together under realistic conditions.

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